R Code

```r
#####################################################################
# PRACTICAL 4: AUTOREGRESSIVE (AR) MODELS USING R

# Objective:
# To understand the structure of autoregressive (AR) models
# and to fit AR models of different orders using R.
# The practical also aims to interpret the estimated
# autoregressive coefficients and relate the AR model
# to the dependence structure studied using ACF and PACF.
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# ============================================================
# 1. INTRODUCTION TO THE DATASET
# ============================================================

# We use the built-in LakeHuron dataset.

data(LakeHuron)

# Display the dataset
LakeHuron



# ============================================================
# 2. BASIC INFORMATION ABOUT THE DATA
# ============================================================

# Class of the data
class(LakeHuron)

# Number of observations
length(LakeHuron)

# Starting time
start(LakeHuron)

# Ending time
end(LakeHuron)

# Frequency of the data
frequency(LakeHuron)



# ============================================================
# 3. PLOT THE TIME SERIES
# ============================================================

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```r
plot(LakeHuron,
     main = "Annual Level of Lake Huron",
     xlab = "Year",
     ylab = "Lake Level")



# ============================================================
# 4. AUTOCORRELATION FUNCTION (ACF)
# ============================================================

acf(LakeHuron,
    main = "ACF of Lake Huron")

# ACF helps us to study the correlation between
# the current observation and its previous observations.



# ============================================================
# 5. PARTIAL AUTOCORRELATION FUNCTION (PACF)
# ============================================================

pacf(LakeHuron,
     main = "PACF of Lake Huron")

# PACF measures the direct relationship between the current
# observation and a lagged observation after accounting for
# the intermediate lags.



# ============================================================
# 6. FITTING AN AR(1) MODEL
# ============================================================

# Fit an exact AR(1) model

ar1 <- ar(LakeHuron,
          order.max = 1,
          aic = FALSE)

ar1



# Extract the AR(1) coefficient

ar1$ar



# Estimated mean of the series

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ar1$x.mean



# Innovation variance

ar1$var.pred



# ============================================================
# 7. OBTAIN THE FITTED AR(1) EQUATION
# ============================================================

# Estimated AR(1) coefficient
phi1 <- ar1$ar[1]

# Estimated mean
mu1 <- ar1$x.mean

# Calculate the constant
c1 <- (1 - phi1) * mu1

# Display estimated values

phi1
mu1
c1


# Display the fitted AR(1) equation

cat("Fitted AR(1) model:\n")

cat("Y(t) =",
    round(c1, 4),
    "+",
    round(phi1, 4),
    "* Y(t-1) + epsilon(t)\n")



# ============================================================
# 8. FITTING AN AR(2) MODEL
# ============================================================

# Fit an exact AR(2) model

ar2 <- ar(LakeHuron,
          order.max = 2,
          aic = FALSE)

ar2

```

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# Extract AR(2) coefficients

ar2$ar



# Estimated mean

ar2$x.mean



# Innovation variance

ar2$var.pred



# ============================================================
# 9. OBTAIN THE FITTED AR(2) EQUATION
# ============================================================

# Extract coefficients

phi1_ar2 <- ar2$ar[1]
phi2_ar2 <- ar2$ar[2]

# Estimated mean

mu2 <- ar2$x.mean

# Calculate the constant

c2 <- (1 - phi1_ar2 - phi2_ar2) * mu2


# Display estimated values

phi1_ar2
phi2_ar2
mu2
c2



# Display the fitted AR(2) equation

cat("Fitted AR(2) model:\n")

cat("Y(t) =",
    round(c2, 4),
    "+",
    round(phi1_ar2, 4),
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    "* Y(t-1) +",
    round(phi2_ar2, 4),
    "* Y(t-2) + epsilon(t)\n")



# ============================================================
# 10. FITTING AN AR(3) MODEL
# ============================================================

# Fit an exact AR(3) model

ar3 <- ar(LakeHuron,
          order.max = 3,
          aic = FALSE)

ar3



# Extract AR(3) coefficients

ar3$ar



# Estimated mean

ar3$x.mean



# Innovation variance

ar3$var.pred



# ============================================================
# 11. OBTAIN THE FITTED AR(3) EQUATION
# ============================================================

# Extract coefficients

phi1_ar3 <- ar3$ar[1]
phi2_ar3 <- ar3$ar[2]
phi3_ar3 <- ar3$ar[3]

# Estimated mean

mu3 <- ar3$x.mean

# Calculate the constant

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```r
c3 <- (1 - phi1_ar3 - phi2_ar3 - phi3_ar3) * mu3


# Display estimated values

phi1_ar3
phi2_ar3
phi3_ar3
mu3
c3


# Display the fitted AR(3) equation

cat("Fitted AR(3) model:\n")

cat("Y(t) =",
    round(c3, 4),
    "+",
    round(phi1_ar3, 4),
    "* Y(t-1) +",
    round(phi2_ar3, 4),
    "* Y(t-2) +",
    round(phi3_ar3, 4),
    "* Y(t-3) + epsilon(t)\n")



# ============================================================
# 12. COMPARISON OF AR COEFFICIENTS
# ============================================================

# AR(1) coefficient
ar1$ar

# AR(2) coefficients
ar2$ar

# AR(3) coefficients
ar3$ar



# ============================================================
# 13. COMPARISON OF ACF AND PACF
# ============================================================

acf(LakeHuron,
    main = "ACF of Lake Huron")

pacf(LakeHuron,
     main = "PACF of Lake Huron")

# The PACF can help us identify a possible order
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# for an autoregressive model.



# ============================================================
# 14. IMPORTANT POINTS
# ============================================================

# AR(1):
# The current observation depends on one previous observation.

# AR(2):
# The current observation depends on two previous observations.

# AR(3):
# The current observation depends on three previous observations.

# The AR coefficients indicate the strength and direction
# of dependence on previous observations.



#####################################################################
# PRACTICE QUESTION
#####################################################################

# Dataset: JohnsonJohnson
#
# JohnsonJohnson contains quarterly earnings per share (EPS)
# data of Johnson & Johnson company.

# Using the built-in JohnsonJohnson dataset:

# (a) Load and display the dataset.
#
# (b) Find its class, length, starting time,
#     ending time and frequency.
#
# (c) Plot the time series.
#
# (d) Plot the ACF and PACF.
#
# (e) Fit an AR(1) model.
#
# (f) Extract the estimated AR coefficient.
#
# (g) Calculate the constant of the fitted AR(1) model.
#
# (h) Write the fitted AR(1) equation by substituting
#     the estimated parameter values.
#
# (i) Interpret the estimated AR coefficient.

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